

Documentation of Features Extracted from the Curves

Total Energy (Energy_J)

Total dissipated energy is fundamental to understanding the thermal input during welding. It represents the actual amount of heat generated in the weld nugget, which is directly linked to material fusion and, consequently, to the mechanical strength (shear strength) of the final weld. Analyzing energy helps assess whether the cycle provided enough heat to ensure a robust joint.

Energy was calculated by integrating electrical power over the welding time. First, instantaneous power was calculated by multiplying the voltage and current vectors point by point. Then, all values of this power curve were summed and multiplied by the fixed time interval between samples (Δt , approximately 39 microseconds).

Maximum Peaks (Vmax, Imax, Rmax)

Peak values show the extremes reached during the process. Maximum resistance, for example, is strongly related to the moment of greatest heating and expansion of the material before it melts and resistance begins to drop. Monitoring these peaks in the curves allows identification of instabilities in the power supply and inference of characteristics of the physical state transition of the metal sheets.

These data were obtained by searching for the highest numerical value within each time series (voltage, current, and dynamic resistance) throughout the entire recorded period, using an absolute maximum (max) mathematical function on the curve data of each sample.

Peak Times (TVmax, TImax, TRmax) and Relative Times

The exact moment at which peaks occur indicates the temporal dynamics of the welding process. Knowing the instant at which resistance reaches its peak helps understand the thermal and mechanical speed of fusion. Peak times that are too delayed or too early relative to the current pulse may indicate problems such as sheet contamination, electrode wear, or poor initial contact.

Absolute times were extracted by locating the exact position (index) on the curve where the respective maximum value occurred and multiplying it by the sampling interval. Relative times (RelativeTVmax, etc.) were calculated by subtracting from the absolute time the instant at which the weld actually began (when power exceeded 10% of the peak), thus normalizing the metric relative to the actual start of the pulse.

Effective Cycle Time (CycleTime)

The real and effective duration of the welding cycle during which significant energy flow occurred often differs slightly from the nominal time programmed on the equipment. Knowing the actual time window during which current actively flowed is crucial for understanding thermal evolution, since short

pulses do not melt the material and excessive pulses can cause violent expulsion of the molten metal.

To calculate this duration, a cutoff threshold equivalent to 10% of the power peak was defined to disregard electrical noise. The start of the cycle was recorded at the first index that exceeded this threshold, and the end at the last index before power permanently dropped again. The effective duration was obtained from the difference between these two positions multiplied by the time resolution (Δt).

Maximum Rate of Change (dV_{Max} , dI_{Max} , dP_{Max} , dR_{Max})

The maximum rate of change, or derivative limit of the curves, indicates how quickly and abruptly electrical conditions change at their most critical event. A sudden, near-instantaneous drop in the resistance curve, for example, is the classic thermal signature of a spatter event (expulsion of molten metal). Analyzing these derivative extremes is one of the most efficient ways to capture anomalous events that weaken the weld.

The calculation was performed by computing the difference between consecutive points (discrete derivative) along each curve using the diff function. This delta was then divided by the continuous time interval (Δt) to reflect the actual variation over time. Finally, the highest value of this instantaneous rate was extracted from the entire record.

RMS Values (V_{rms} , I_{rms} , P_{rms})

The RMS value (Root Mean Square) is the most appropriate statistical measure for representing the true ("effective") magnitude of pulsed or alternating current waveforms. Unlike a simple average, it precisely quantifies the wave's real continuous capacity to perform electrical work and provide heating to the weld.

The RMS of each quantity (voltage, current, and power) was calculated by first squaring all individual points of the curve, taking the mathematical average of all these resulting values within the pulse, and finally applying the square root to the final average. This was done using the native rms function across all observed samples.

Total Electric Charge (Charge_C_)

Electric charge reflects the absolute accumulated amount of electrons driven through the sheets during the entire welding event. When analyzed together with energy and RMS profiles, deviations in total charge can expose equipment impedances and oscillations that might not otherwise be easily noticed, in addition to serving as a conservative measure of overall circuit stress.

Charge in Coulombs was generated by performing the time integral of the electric current curve end-to-end. Computationally, this represented the simple sum of all current intensity values captured along the curve, followed by multiplication by the fixed sampling time step size in seconds (Δt).